

Outlier Removal: Z-Score Method Explained

1. The Core Logic: Cholesterol Filtering

Line: `df1 = df[df.Cholesterol <= (df.Cholesterol.mean() + 3*df.Cholesterol.std())]`

This keeps rows where Cholesterol $\leq$ Mean + 3 * Standard Deviation.

Why 3 * std?

Based on the Empirical Rule (68-95-99.7), 99.7% of data in a normal distribution falls within 3 standard deviations. Anything beyond this is statistically treated as an extreme outlier.

2. Checking for High Extremes (MaxHR & FastingBS)

When checking MaxHR and FastingBS using the +3 std threshold, if the resulting dataframe is empty, it means there are no extreme outliers on the upper side of the distribution for these specific features.

3. Sequential Removal (Oldpeak & RestingBP)

For Oldpeak and RestingBP, outliers were identified and removed sequentially:

- Oldpeak: Reduced rows from 915 to 909 (6 rows removed).
- RestingBP: Reduced rows further to 902.

This cleaning ensures that extreme values (like BP of 190 or 200) do not skew the model's perception of 'normal'.

4. Impact on Machine Learning Models

Outliers can skew mean/std, distort linear models, and increase variance. Removing them is critical for algorithms like Logistic Regression, Linear Regression, and SVM.

Note: This specific implementation is a 'Half Z-Score' filter as it only targets the upper tail (High values). A full Z-score method would use: $|(x - \text{mean}) / \text{std}| > 3$ to catch both extremes.

5. Conclusion & Benefits

Final Dataset: Shrinks to 902 rows.

Benefits:

- Improved Model Robustness
- Higher Statistical Validity

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- Better Training Stability